BRIAN BEAUMONT AVANTS, Ph.D.

AI & Machine Learning Executive · Computational Anatomy Pioneer · Founder, Cingulate LLC

Cingulate LLC, 145 High Street, Portsmouth, NH, USA +1 (415) 508-4437 antsx@antsx.com stnava.github.io
github.com/stnava · github.com/ANTsX linkedin.com/in/brian-avants ORCID: 0000-0002-4212-3362
Google Scholar: **57,000+ Citations** · **h-index: 71** · **i10-index: 165** PubMed Author Profile

EXECUTIVE & SCIENTIFIC SUMMARY: Entrepreneurial AI/ML executive, principal data scientist, and computational imaging leader with over two decades of impact spanning academic medicine, pharmaceutical drug development, clinical trial CROs, and open-source software engineering. Co-creator and lead architect of the **ANTsX ecosystem** (`ANTs`, `ANTsPy`, `ANTsR`, `ANTsPyNet`, `ANTsPyMM`), the international standard for diffeomorphic medical image registration (`SyN`), morphometry, and deep learning phenotyping (**57,000+ citations, h-index 71**). Proven executive track record founding **Cingulate LLC**, directing global Machine Learning teams at **Invicro**, and leading translational imaging biomarker strategy for the landmark *Aducanumab/Aduhelm* program at **Biogen**.

57,000+ Citations	**h-index: 71 / i10: 165**	**6x #1 Global Benchmarks**	**Pharma & Venture AI Lead**
Foundational imaging & ML methods	180+ peer-reviewed publications	Klein, EMPIRE10, BRATS, SATA	Cingulate, Invicro, Biogen, UVA, UPenn

ENTREPRENEURIAL, EXECUTIVE & ACADEMIC APPOINTMENTS

Founder & Principal Consultant — 2018 – Present
Cingulate LLC — Portsmouth, NH

- Founder and managing principal of an independent AI, machine learning, and computational neuroimaging consultancy serving biotechnology, pharmaceutical, defense (DoD/CDMRP), and academic research organizations.
- Architect of novel interpretable representation learning and matrix optimization tools, including **NSA-Flow** (*Non-Negative Stiefel Approximating Flow*, PyTorch) and **SiMLR** (*Similarity-driven Multi-view Linear Reconstruction*, *Nature Computational Science* 2021).
- Lead multi-modal neuroimaging phenotyping, longitudinal subtyping, and normative database engineering for large-scale studies including the **UK Biobank**, **Parkinson's Progression Markers Initiative (PPMI)**, and **LIMBIC-CENC**.

Visiting Associate Professor of Radiology and Medical Imaging — 2018 – Present
University of Virginia School of Medicine — Charlottesville, VA

- Co-lead multi-view machine learning and quantitative neuroimaging investigations of repetitive low-level blast exposure and traumatic brain injury (TBI) in military Special Operations personnel and breachers (*Journal of Neurotrauma* 2020, 2022, 2024).
- Co-architect of the **Normative Neuroimaging Library** (*J. Neurotrauma* 2024) and **Deep Computational Anatomy via Latent-Aligned Multiview Normalizing Flows** (2026).

Senior Director of Machine Learning — 2018 – 2024
Invicro (a Konica Minolta Company) — Boston, MA

- Directed the enterprise Machine Learning group for a premier global imaging CRO, leading clinical trial software development and integrative AI analytics across neurodegenerative, oncology, cardiopulmonary, and multi-species drug development programs.
- Designed and deployed **ANTsPyMM**, automated deep learning quality control, 3D concurrent super-resolution, and multi-modal patient representation learning pipelines (*ICLR* 2023, *Patterns* 2023, *European Journal of Human Genetics* 2024).

Visiting Scholar, Institute for Computational and Experimental Research in Mathematics (ICERM) — Spring 2020
Brown University — Providence, RI

- Invited visiting scientist for the semester program *"Model and Dimension Reduction in Uncertain and Dynamic Systems"*; developed mathematical foundations for integrative multi-view machine learning (`SiMLR`).

Associate Director, Translational & Advanced Analytics (Imaging Biomarkers) — 2016 – 2018
Biogen — Cambridge, MA

- Lead integrative imaging data scientist for the **Aducanumab (Aduhelm)** clinical trial program (Phase 1b PRIME and Phase 3 ENGAGE/EMERGE); performed confirmatory amyloid PET SUVR analyses and created Figure 1 and the cover visualization for the landmark **2016 *Nature* publication**.
- Led longitudinal network-specific rs-fMRI biomarker discovery in amyloid-positive cohorts (*Neurobiology of Aging* 2018) and initiated **ANTsPy** development for deep learning in neuroimaging.

Assistant Professor of Radiology — 2012 – 2016
University of Pennsylvania Perelman School of Medicine (PICSL) — Philadelphia, PA

- Directed computational neuroimaging, multi-modal MRI/DTI/PET/ASL methodology, and big-data imaging genetics projects; served as Principal Investigator on NIH BD2K K01 Career Development Award and Image Analysis Lead for IARPA KRNS.

Research Associate (2008–2012) · Postdoctoral Research Fellow (2006–2008) — 2006 – 2012
University of Pennsylvania, Department of Radiology — Philadelphia, PA

- Developed foundational methods for **Symmetric Diffeomorphic Image Registration (`SyN`)**, **N4ITK** bias correction, **Atropos** multivariate segmentation, **DiReCT** cortical thickness, and **Sparse Canonical Correlation Analysis (`SCCAN`)**.

Algorithm Developer, Imaging Department — **1999 – 2000**

Siemens Corporate Research — Princeton, NJ

- Developed adaptive minimal-path vessel tracking and 3D segmentation algorithms in CTA/CE-MRA volumetric imaging.

EDUCATION & POSTDOCTORAL TRAINING

2006 – 2008 — **Postdoctoral Fellowship in Radiology**, University of Pennsylvania, Philadelphia, PA.

2005 — **Ph.D. in Bioengineering**, University of Pennsylvania, Philadelphia, PA. Dissertation: *Diffeomorphometry for Computational Anatomy*. NIH Institute for Medicine and Engineering Trainee (2002).

2002 / 2003 — **M.S. in Computer Science**, University of Pennsylvania, Philadelphia, PA.

1998 — **B.A. in Physics and Literature**, New College of Florida, Sarasota, FL (*With High Honors*; National Merit Scholar).

RESEARCH DOMAINS & TECHNICAL EXPERTISE

Deep Learning & AI — High-dimensional multi-output regression, 3D segmentation, concurrent super-resolution, longitudinal manifold embedding (Aligned-UMAP), Stiefel manifold flow (`NSA-Flow`), normalizing flows, multi-modal prediction with missing modalities, and interpretable neural architectures.

Brain Mapping — Brain structure, function, perfusion, neuroinflammation, and cognition across development, aging, neurotrauma, and neurodegeneration; multi-view modality integration (`SiMLR`, `SCCAN`); disease subtyping and staging (`subtyper`); reproducible biobank-scale phenotyping.

Tech Stack — C++, Python, PyTorch, TensorFlow/Keras, R, Julia, CMake, ITK, Bash, Git, LaTeX, Docker/Singularity, AWS/HPC cloud architectures.

FLAGSHIP SOFTWARE CONTRIBUTIONS & OPEN-SOURCE ECOSYSTEM

ANTs (2008–) — Advanced Normalization Tools (github.com/ANTsX/ANTs): C++ library used globally across academia and industry for medical image registration (`SyN`), segmentation (`Atropos`), and cortical thickness; **Creator & Lead Architect**.

ANTsPy & ANTsPyNet — Python interface and deep learning library (ANTsPy, ANTsPyNet) providing pre-trained 2D/3D networks for brain extraction, Deep Atropos, HippMapp3r, brain age, lesion segmentation, and super-resolution; **Co-Creator & Architect**.

ANTsPyMM (2021–) — Multi-modal neuroimaging processing and automated QC framework (ANTsX/ANTsPyMM) converting structural T1w, FLAIR, rs-fMRI, DTI, and neuromelanin MRI into standardized Image Derived Phenotypes (IDPs); **Lead Architect**.

ANTsR (2012–) — R statistical programming extension of ANTs (ANTsX/ANTsR) featuring `Eigenanatomy`, `SCCAN`, and `SiMLR`; **Creator & Lead Designer**.

NSA-Flow (2025–) — PyTorch package (stnava/nsa_flow) implementing Non-Negative Stiefel Approximating Flow for orthogonalish, sparse, interpretable matrix factorization and representation learning; **Creator**.

subtyper (2023–) — Disease subtyping, staging, and latent space evaluation toolkit (stnava/subtyper); **Creator**.

ITKv4 (2002–) — The Insight ToolKit (itk.org): **Lead Designer** for the v4 deformable image registration framework.

Neuroconductor — R platform for reproducible neuroimaging data science (neuroconductor.org); **Co-Founder & Contributor**.

AWARDS, INTERNATIONAL BENCHMARKS & HONORS

2014 — **Best Paper Award** (with Nicholas J. Tustison) at the STACOM 2014 Challenge, MICCAI.

2013 — **1st Place, BRATS 2013 Brain Tumor Segmentation Challenge**, MICCAI, Nagoya, Japan (`ANTsR`, with N. Tustison).

2013 — **1st Place, SATA Segmentation Challenge**, MICCAI, Nagoya, Japan (`ANTs`, with N. Tustison).

2012–2013 — **"Standard Setting" Image Registration Software** selected for the MICCAI 2013 Multi-Atlas Segmentation Challenge (4D cardiac, brain, and multimodality canine leg MR images); all top finishers in MICCAI 2012 multi-atlas segmentation used `ANTs`.

2010 — **1st Overall** in the unbiased **EMPIRE10** International Thoracic CT Lung Registration Challenge (*IEEE TMI* 2011).

2009 — **1st Rank** in the unbiased **Klein 2009** International Brain MRI Registration Evaluation of 14 nonlinear algorithms (*NeuroImage* 2009).

2007 — **Young Investigator Award**, General Medical Computing, MICCAI, Brisbane, Australia.

2003 **Best Poster Award**, SPIE Medical Imaging.

1995 **Future Healthcare Researcher Scholarship**, Humana Health Care.

1994 **National Merit Scholarship**, New College of Florida, Sarasota, FL.

GRANTS, SPONSORED RESEARCH & CONSORTIUM LEADERSHIP

2018 – Present **DoD** / **CDMRP** / **LIMBIC-CENC Neurotrauma & Blast Overpressure Research Programs** (with James R. Stone, M.D., Ph.D. & Nicholas J. Tustison, D.Sc., University of Virginia): Multi-view neuroimaging biomarkers (`SiMLR`, `NSA-Flow`), TSPO PET neuroinflammation, and Normative Neuroimaging Library development in Special Operations personnel and combat breachers (Role: **Co-Investigator** / **Lead ML Consultant**).

2018 – 2024 **Parkinson's Progression Markers Initiative (PPMI) & Multimodal Neurodegeneration Consortia** (Invicro / Michael J. Fox Foundation / NIH CARD Collaborations): Automated multi-modal MRI/PET phenotyping (`ANTsPyMM`), Aligned-UMAP longitudinal subtyping, and missing-modality patient representation learning (Role: **Senior Director of ML** / **Co-Investigator**).

10/2014 – 09/2019 **The Imaging Genomics of Pediatric Executive Function**, NIH Big Data to Knowledge (BD2K), **K01-ES025432-01** (**Avants, Brian, PI**), $275,000/annual direct costs, 80% effort.

08/2015 – 05/2018 **Multimodal Brain Maturation Indices Modulating Psychopathology and Neurocognition**, NIH **R01-MH107235-01** (Gur, Ruben, PI), $275,000/annual direct costs (Role: **Collaborator**).

09/2013 – 08/2018 **Neuroscience Neuroimaging Center**, NIH **P30-NS045839-07** (Detre, John, M.D., PI), $499,727/annual direct costs (Role: **Collaborator** / **Technical Core Lead**).

2014 **Knowledge Representation in Neural Systems (KRNS)**, IARPA-BAA-12-05, 50% effort (Role: **Image Analysis Lead**).

09/2012 – 07/2013 **Age, Hearing Loss, and Sentence Comprehension: Neural Correlates**, NIA/NIH **5-R01-AG038490-05** (Grossman, Murray, PI), $291,978/annual direct costs (Role: **Co-PI**).

09/2011 – 08/2015 **Continued Development and Maintenance of ITK-SNAP 3D Image Segmentation Software**, NIH **R01-EB014346-01** (Yushkevich, Paul A., PI), $350,824/annual direct costs (Role: **Collaborator**).

09/2010 – 08/2015 **TDP-43 Proteinopathies in ALS-Dementia**, NIH **P01-AG032953-02** (Lee, Virginia / Grossman, Murray, PI), $729,272/annual direct costs (Role: **Collaborator**).

06/2010 – 06/2013 **Fundamental Refactoring of Deformable Image Registration in ITK with Distributed Computing and GPU Acceleration**, National Library of Medicine **HHSN276201000492P** (Gee, James C., PI), $472,861/annual direct costs (Role: **Co-PI & Architecture Lead**).

02/2011 – 01/2012 **Longitudinal Multi-Modal Neuroimaging of Natural Recovery after Traumatic Brain Injury: A Pilot Study**, Moss Rehab Research Institute / NIH (**Avants, Brian, PI**), $7,764/annual direct costs.

08/2007 – 04/2013 **In Utero Cocaine Exposure: Adolescent & Young Adult Neurocognitive Outcome**, CHOP / NIH **R01-DA14129-08** (Gee, James C., PI), $102,515/annual direct costs (Role: **Collaborator**).

06/2010 – 05/2012 **Parkinson's Disease & Dementia**, NIH **P50-NS053488-04S1** (Trojanowski / Grossman, PI), $132,641/annual direct costs (Role: **Collaborator**).

09/2009 – 08/2012 **Tract-Specific Analysis of Brain White Matter**, NIH **R01-NS065347-02** (Gee, James C., PI), $370,667/annual direct costs (Role: **Collaborator**).

09/2009 – 05/2010 **Pediatric Template of Brain Perfusion**, NIMH/NIH **3-R01-MH080892-01A2S1** (Wang, Jiongjiong, PI) (Role: **Co-PI**).

09/2008 – 09/2011 **Advanced Neuroimage Registration Methods: Effects of Prenatal Cocaine Exposure**, NIH (Gee, James C., PI; **Avants, Brian, Co-PI**), $354,375/annual direct costs.

06/2008 – 01/2011 **The Longitudinal Effect of Neurodegeneration on Language-Network Neuroanatomy and Cognition**, UPenn Institute on Aging (Gee, James C., PI; **Avants, Brian, Co-PI**).

04/2007 – 01/2011 **Long Term Effects of Prenatal Cocaine Exposure**, NIH (Hurt, Hallam, PI) (Role: **Research Scientist**).

04/2006 – 03/2010 **Shape Optimizing Diffeomorphisms for Computational Biology**, NIH / UCLA Center for Computational Biology **R01-EB006266** (Gee, James C., PI; based on B. Avants' Ph.D. thesis).

08/2004 – 06/2009 **Surface-based Cortical Analysis in ITK: Segmentation, Conformal Flattening and Statistics**, NLM-NIH ITK Development Grant (Gee, James C., PI; **Avants, Brian, Co-PI**).

09/2002 – 06/2005 **Bioengineering Training in Cardiovascular Pathophysiology**, Institute for Medicine and Engineering Training Grant, UPenn (Davies, Peter, PI; **Trainee**).

EDITORIAL POSITIONS, ADVISORY PANELS & PROFESSIONAL SERVICE

2019 **Brain Injury Biomarker Strategy Panel Contributor**, sponsored by Cohen Veterans Bioscience.

2017	**Topic Editor**, *NeuroImage* Special Issue on Brain Parcellation.
2012 – 2015	**Topic Editor**, *Frontiers in Neuroinformatics*: Research Topic on Neuroinformatics with the Insight ToolKit (ITK).
2013 – Present	**Grant Reviewer**, Alzheimer's Association & Ad Hoc Federal/International Study Sections.
2013 – Present	**Consultant**, American College of Radiology (ACR) Head Injury Institute Information Technology Committee.
2010 – Present	**Insight ToolKit (ITKv4) Development Leader**, NLM/NIH-sponsored industry-academic consortium.
2011 – Present	**Task Force Member**, International Neuroinformatics Coordinating Facility (INCF) Registration and Atlasing Task Force.
2006 – Present	**Member**, Neuroimaging Informatics Tools and Resources Clearinghouse (NITRC) & Spatial Transformations Informatics Technology Initiative (SIFTI).
Peer Review	**Ad Hoc Referee:** *Nature Neuroscience*, *Nature Computational Science*, *Nature Methods*, *Medical Image Analysis*, *NeuroImage*, *IEEE Transactions on Medical Imaging*, *IEEE TPAMI*, *Human Brain Mapping*, *Medical Physics*, *Pediatrics*.
Memberships	Organization for Human Brain Mapping (OHBM); International Society for Magnetic Resonance in Medicine (ISMRM).

MENTORSHIP, ADVISING & MAJOR RESEARCH COLLABORATIONS

2018 – Present	**Industry & Consortium Mentorship (Invicro, Cingulate LLC, UVA):** Mentored machine learning scientists, postdoctoral fellows, and Ph.D. interns across deep learning, multimodal representation learning, and longitudinal clinical trial modeling (including collaborations with Anant Dadu, Zhenbang Wu, Brandon Theodorou, and Taylor Gosselin).
Summer 2017	**Industry Internship Advisor to Nicholas Cullen** (Biogen Biomarkers Group): Deep learning and multivariate neuroimaging in Alzheimer's Disease; co-created `ANTsPy` and co-authored Springer Protocols chapter on deep learning brain segmentation.
2014 – 2016	**Faculty Technical Advisor to Dr. Dorian Pustina** (UPenn Postdoctoral Scientist, with Dr. H. Branch Coslett): Automated stroke lesion segmentation (`LINDA`) and multivariate lesion-symptom mapping (`SCCAN`); *subsequently Director of Experimental Neuroimaging at CHDI Foundation*.
2012 – 2017	**Ph.D. Co-Advisor to Dr. Efstathios (Stathis) Gennatas** (UPenn Neuroscience): Developmental brain morphometry across childhood to young adulthood (*J. Neurosci.* 2017); *subsequently faculty/scientist at UCSF*.
2010 – 2015	**Ph.D. Advisor to Dr. Benjamin M. Kandel** (UPenn Bioengineering): Eigenanatomy, sparse linear regression, and ASL perfusion decomposition (Oral presentation at IPMI 2013).
2010 – 2015	**Ph.D. Co-Advisor to Dr. Paramveer S. Dhillon** (UPenn Computer Science, with Prof. Lyle Ungar): Prior-based Eigenanatomy and functional parcellation (Oral presentation at PRNI 2013; *subsequently MIT postdoc and faculty at University of Michigan*).
2012 – 2013	**Organizer, UPenn Provost's Interdisciplinary Seminar Series on Multi-Modality Brain Mapping** (*Brainomics*): Hosted David Van Essen (Human Connectome Project), Mert Sabuncu, and Moriah Thomason.
Key Collaborations	**Dr. James R. Stone & Dr. Nicholas J. Tustison** (UVA/UC Irvine: ANTsX, SiMLR, NSA-Flow, military blast neurotrauma); **Dr. Mike Nalls & Dr. Faraz Faghri** (NIH CARD / Data Tecnica: multimodal AI & Parkinson's subtyping); **Dr. Murat Maga** (Seattle Children's/UW: evolutionary morphometrics & `SlicerMorph`); **Dr. Murray Grossman** (UPenn FTLD Center); **Dr. Martha Farah & Dr. Hallam Hurt** (UPenn/CHOP socioeconomic brain development); **Dr. Ruben Gur** (UPenn Brain-Behavior Lab); **Dr. Howie Rosen** (UCSF Neurology); **Dr. Yvette Sheline** (UPenn CNDS).

HIGHLIGHTED MEDIA & COVER FEATURES

2016	***Nature* Cover Story (Aducanumab PRIME Study):** Performed confirmatory amyloid PET analysis and created Figure 1 and the cover visualization for Sevigny et al., *"The antibody aducanumab reduces Aβ plaques in Alzheimer's disease"* (*Nature* 537, 50–56).
2013	**CNN Health Feature:** Coverage of collaborative neuroimaging research with Dr. Martha Farah on childhood environment and brain development.
2012	***The Washington Post* & *Daily Mail* (UK):** International press coverage of Society for Neuroscience (SfN) press conference presentation on early home environment and adolescent cortical thickness.
2012	**Penn Medicine News:** Feature article (*"Image Wizardry"*) highlighting `ANTs` winning the MICCAI 2012 segmentation competition.

SELECTED BRAIN MAPPING & DATA SCIENCE TUTORIALS

May 2019	"Introduction to the ANTsX Ecosystem," Workshop Tutorial, ICERM, Brown University, Providence, RI.
Jun 2018	"ANTsX for Decoding and Resting-State fMRI Studies," Talk & Tutorial, Haxby Lab, Dartmouth College, Hanover, NH.

Aug 2017	"ANTs and ANTsR for Clinical Applications," Two-Day Tutorial, Seattle Children's / University of Washington, Seattle, WA.
Aug 2016	"ANTs and ANTsR in Biostatistics," Two-Day Tutorial, MD Anderson Cancer Center, Houston, TX.
Jul 2016	Data Science Domain Expert Invitee, Park City Mathematics Institute (PCMI), Park City, UT.
Jun 2015	"ANTs and ANTsR," One-Day Tutorial, USC Center for Imaging Genetics (ENIGMA), Los Angeles, CA.
May 2015	"ANTs and ANTsR in Machine Learning," Two-Day Tutorial (with N. Tustison), McGill University & Montreal Neurological Institute, Montreal, QC.
Sep 2013	"Multivariate Medical Imaging Analysis with R," MICCAI 2013 Tutorial, Nagoya, Japan.
Jul 2013	"ANTs & Eigenanatomy," UCLA Neuroimaging Training Program, Los Angeles, CA.
Sep 2011	"ITKv4 Image Registration Tutorial" (with Kitware, Inc.), MICCAI 2011, Toronto, ON.
Oct 2006	"Tutorial on Neuromorphometry," MICCAI 2006, Copenhagen, Denmark.

SELECTED INVITED LECTURES & KEYNOTES

2018 – 2025	**Invited Industry, Consortium & Academic Presentations:** Michael J. Fox Foundation (MJFF) PPMI Data Science Working Group; LIMBIC-CENC Annual Conferences; ICERM Workshop on Dimension Reduction (Brown University); Invicro Translational Biomarker Symposia.
Jun 2017	"Translational Brain Mapping in Neurodegenerative Disease," Yale University, New Haven, CT.
Apr 2017	Platform Presentation on Quantitative Amyloid PET & Connectivity Biomarkers, AD/PD 2017, Vienna, Austria.
Mar 2017	"Multiple Modality Analysis of Alzheimer's Disease," Duke University, Durham, NC.
Jun 2015	"Large-Scale, Predictive Analytics for M^3 Pediatric Neuroimaging," ISMRM Annual Meeting, Toronto, ON.
May 2015	"Introduction to the ANTs Ecosystem," University of California, San Diego, La Jolla, CA.
Nov 2014	"Functional Imaging Analysis with R," University of South Carolina, Columbia, SC.
Aug 2014	"Big Image Registration," Frontiers in Human Brain Mapping Workshop, Princeton University, Princeton, NJ.
Apr 2014	"Eigenwords & Eigenanatomy for Decoding Neural Representations of Semantics," HRL Laboratories, Malibu, CA.
Feb 2014	"ANTs & Eigenanatomy for Integrative Brain Mapping," University of Calgary, Calgary, AB.
Oct 2012	"A Unified Image Registration Framework for ITKv4," National Library of Medicine (NIH), Bethesda, MD.
Oct 2012	"Early Home Environment Impacts Cortical Thickness in Young Adulthood," SfN Press Conference, New Orleans, LA.
Jun 2012	"Eigenanatomy Methods for Multivariate Brain Mapping," University of California, San Francisco, CA.
May 2012	"Early Home Environment Impacts Cortical Thickness," Chinese Academy of Sciences, Beijing, China; & Incheon National University, South Korea.
2011	Invited Lectures at UCLA (Dec 2011), Society for Neuroscience (Nov 2011), and Washington University in St. Louis (Jan & Feb 2011).
2005 – 2010	Invited Lectures at Chinese Academy of Sciences (2010), National Library of Medicine (2010), Columbia University (2009), SRI International (2009), Stanford University (2008), HHMI Janelia Farm (2008), ISBI (2006, 2007), and ISI Utrecht (2005).

PEER-REVIEWED JOURNAL PUBLICATIONS & RECENT PREPRINTS (1999 – 2026)

Full citation metrics (**57,000+ citations, h-index: 71**): Google Scholar Profile · PubMed Index

1. Tustison, N. J.; **Avants, B. B.**; Cook, P. A.; Gee, J. C.; Stone, J. R. (2026). *Deep Computational Anatomy via Latent-Aligned Multiview Normalizing Flows.* bioRxiv / PubMed (PMID: 42146611).

2. Theodorou, B.; Dadu, A.; **Avants, B.**; Nalls, M.; Sun, J.; Faghri, F. (2026). *Realistic PET image synthesis from MRI for automated inference of brain atrophy and Alzheimer's*. **iScience**, 29(5), 115747.
3. **Avants, B. B.**, et al. (2025). *Non-Negative Stiefel Approximating Flow: Orthogonalish Matrix Optimization for Interpretable Embeddings*. **arXiv**:2511.06425.
4. **Avants, B. B.**; Tustison, N. J.; Stone, J. R. (2025). *Joint representations from multi-view MRI-based learning support cognitive and functional performance domains*. **medRxiv**.
5. Tustison, N. J.; Yassa, M. A.; Rizvi, B.; Cook, P. A.; Holbrook, A. J.; Sathishkumar, M. T.; Tustison, M. G.; Gee, J. C.; Stone, J. R.; **Avants, B. B.** (2024). *ANTsX neuroimaging-derived structural phenotypes of UK Biobank*. **Scientific Reports**, 14(1), 8848.
6. Stone, J. R.; **Avants, B. B.**; Tustison, N. J.; Gill, J.; Wilde, E. A.; Neumann, K. D.; Gladney, L. A.; Kilgore, M. O.; Bowling, F.; Wilson, C. M. (2024). *Neurological effects of repeated blast exposure in Special Operations personnel*. **Journal of Neurotrauma**, 41(7–8), 942–956.
7. Tustison, N.; **Avants, B.**; Whitlow, C. T.; Lancashire, L.; Bhatt, S. D.; Haas, M. (2024). *Normative Neuroimaging Library: Designing a Comprehensive and Demographically Diverse Dataset of Healthy Controls to Support Traumatic Brain Injury Diagnostic and Therapeutic Development*. **Journal of Neurotrauma**, 41(23–24), 2497–2512.
8. Yang, Z.; Wen, J.; Erus, G.; **Avants, B. B.**, et al. (2024). *Brain aging patterns in a large and diverse cohort of 49,482 individuals*. **Nature Medicine**, 30, 3015–3026.
9. **Avants, B. B.**; Fonville, L.; Hampton, O.; Reardon, A.; Stenger, A.; Wang, X.; Tustison, N. J.; Stone, J. R.; Cook, P. A.; Marebwa, B.; Chahine, L. M. (2024). *Magnetic Resonance Imaging Data Phenotypes for the Parkinson's Progression Markers Initiative*. **medRxiv**.
10. Dennis, E. L.; Rowland, J.; Esopenko, C.; Tustison, N.; Newsome, M.; **Avants, B.**; Gill, J.; Hinds, S.; Kenney, K.; Lindsey, H. (2024). *Brain volume changes following blast-related mild TBI in service members and veterans: a LIMBIC-CENC study*. **medRxiv**, 2024.02.27.24303460.
11. Polfus, L.; Shirazi, H. M.; Reardon, A.; Varga, M.; Tchourbanov, A.; Gosselin, T.; Dadu, A.; **Avants, B.**; Faghri, F.; Nalls, M.; Hesterman, J. (2024). *Genetic and brain imaging phenotype joint prediction of longitudinal Parkinson's Disease subtypes*. **European Journal of Human Genetics**, 32, 653.
12. Wu, Z.; Dadu, A.; Tustison, N.; **Avants, B.**; Nalls, M.; Sun, J.; Faghri, F. (2023). *Multimodal Patient Representation Learning with Missing Modalities and Labels*. **The Twelfth International Conference on Learning Representations (ICLR)**.
13. Dadu, A.; Satone, V. K.; Kaur, R.; Koretsky, M. J.; Iwaki, H.; Qi, Y. A.; Ramos, D. M.; **Avants, B.**; Hesterman, J.; Gunn, R.; Nalls, M.; Faghri, F. (2023). *Application of Aligned-UMAP to longitudinal biomedical studies*. **Patterns** (Cell Press), 4(6), 100741.
14. **Avants, B.**; Tustison, N. J.; McMillan, C. T.; Gosselin, T.; Gunn, R.; Hesterman, J. (2023). *Concurrent 3D super resolution on intensity and segmentation maps improves detection of structural effects in neurodegenerative disease*. **medRxiv**, 2023.02.02.23285376.
15. Elhamod, M.; Diamond, K. M.; Maga, A. M.; Bakis, Y.; Bart Jr, H. L.; Mabee, P.; Dahdul, W.; Leipzig, J.; Greenberg, J.; **Avants, B.** (2022). *Hierarchy-guided neural network for species classification*. **Methods in Ecology and Evolution**, 13(3), 642–652.
16. **Avants, B. B.**; Tustison, N. J. (2022). *Mapping the Spatial Distribution of Lesions in Stroke: Effect of Diffeomorphic Registration Strategy in the ATLAS Dataset*. In **Lesion-to-Symptom Mapping: Principles and Tools** (pp. 79–94), Springer.
17. Stone, J.; **Avants, B.**; Tustison, N.; Gill, J.; Wilde, E.; Neumann, K.; Gladney, L.; Kilgore, M.; Modica, L. T.; Bowling, F. (2022). *Increased Neuroinflammation in special operators with a history of blast overpressure exposure*. **Journal of Neurotrauma**, 39(11–12), A27–A28.
18. **Avants, B. B.**; Tustison, N. J.; Stone, J. R. (2021). *Similarity-driven multi-view embeddings from high-dimensional biomedical data*. **Nature Computational Science**, 1(2), 143–152.
19. Tustison, N. J.; Cook, P. A.; Holbrook, A. J.; Johnson, H. J.; Muschelli, J.; Devenyi, G. A.; Duda, J. T.; Das, S. R.; Cullen, N. C.; Gillen, D. L.; Yassa, M. A.; Stone, J. R.; Gee, J. C.; **Avants, B. B.** (2021). *The ANTsX ecosystem for quantitative biological and medical imaging*. **Scientific Reports**, 11(1), 9068.
20. Tustison, N. J.; Altes, T. A.; Qing, K.; He, M.; Miller, G. W.; **Avants, B. B.**; Shim, Y. M.; Gee, J. C.; Mugler III, J. P.; Mata, J. F. (2021). *Image-versus histogram-based considerations in semantic segmentation of pulmonary hyperpolarized gas images*. **Magnetic Resonance in Medicine**, 86(5), 2822–2836.
21. Diamond, K. M.; **Avants, B. B.**; Maga, A. M. (2021). *Machine learning-based segmentation and landmarking of 2D fish images*. **Integrative and Comparative Biology**, 61, E1100–E1101.
22. Stone, J. R.; **Avants, B. B.**; Tustison, N. J.; Wassermann, E. M.; Gill, J.; Polejaeva, E.; Dell, K. C.; Carr, W.; Yarnell, A. M.; LoPresti, M. L. (2020). *Functional and structural neuroimaging correlates of repetitive low-level blast exposure in career breachers*. **Journal of Neurotrauma**, 37(23), 2468–2481.
23. Tustison, N. J.; Holbrook, A. J.; **Avants, B. B.**; Roberts, J. M.; Cook, P. A.; Reagh, Z. M.; Duda, J. T.; Stone, J. R.; Gillen, D. L.; Yassa, M. A. (2019). *Longitudinal Mapping of Cortical Thickness Measurements: An ANTs Longitudinal Cortical Thickness Pipeline*. **NeuroImage**, 199, 523–536.
24. **Avants, B. B.**; Hutchison, R. M.; Mikulskis, A.; Salinas-Valenzuela, C.; Hargreaves, R.; Beaver, J.; Chiao, P.; ADNI (2018). *Amyloid beta–positive subjects exhibit longitudinal network-specific reductions in spontaneous brain activity*. **Neurobiology of Aging**, 74, 191–201.
25. Chiao, P.; Bedell, B. J.; **Avants, B. B.**; Zijdenbos, A. P.; Grand'Maison, M.; O'Neill, P.; O'Gorman, J.; Chen, T.; Koeppe, R. (2018). *Impact of Reference/Target Region Selection on Amyloid PET Standard Uptake Value Ratios in the Phase 1b PRIME Study of Aducanumab*. **Journal of Nuclear Medicine**, 60(1), 100–106.
26. Muschelli, J.; Fortin, J. P.; Gherman, A.; **Avants, B. B.**; Whitcher, B.; Caffo, B.; Crainiceanu, C. (2018). *Neuroconductor: An R platform for medical imaging analysis*. **Biostatistics**, 20(2), 218–239.
27. Pustina, D.; **Avants, B. B.**; Faseyitan, O. K.; Medaglia, J. D.; Coslett, H. B. (2018). *Improved accuracy of lesion to symptom mapping with multivariate sparse canonical correlations*. **Neuropsychologia**, 115, 154–166.
28. Pustina, D.; Coslett, H. B.; Ungar, L.; Faseyitan, O. K.; Medaglia, J. D.; **Avants, B. B.**; Schwartz, M. F. (2017). *Enhanced estimations of post-stroke aphasia severity using stacked multimodal predictions*. **Human Brain Mapping**, 38(11), 5603–5615.
29. Maga, A. M.; Tustison, N. J.; **Avants, B. B.** (2017). *A population level atlas of Mus musculus craniofacial skeleton and automated image-based shape analysis*. **Journal of Anatomy**, 231(3), 433–443.
30. Gennatas, E. D.; **Avants, B. B.**; Wolf, D. H.; Satterthwaite, T. D.; Ruparel, K.; Ciric, R.; Hakonarson, H.; Gur, R. E.; Gur, R. C. (2017). *Age-related effects and sex differences in gray matter density, volume, mass, and cortical thickness from childhood to young adulthood*. **Journal of Neuroscience**, 37(20), 5065–5073.
31. Xie, L.; Pluta, J. B.; Das, S. R.; Wisse, L. E. M.; Wang, H.; Mancuso, L.; Kliot, D.; **Avants, B. B.**; Ding, S.; Manjón, J. V.; Yushkevich, P. A. (2017). *Multi-template analysis of human perirhinal cortex in brain MRI: Explicitly accounting for anatomical variability*. **NeuroImage**, 144, 183–202.

32. Olm, C. A.; Kandel, B. M.; **Avants, B. B.**; Detre, J. A.; Gee, J. C.; Grossman, M.; McMillan, C. T. (2016). *Arterial spin labeling perfusion predicts longitudinal decline in semantic variant primary progressive aphasia*. **Journal of Neurology**, 263(10), 1927–1938.
33. Kandel, B. M.; **Avants, B. B.**; Gee, J. C.; McMillan, C. T.; Erus, G.; Doshi, J.; Davatzikos, C.; Wolk, D. A. (2016). *White matter hyperintensities are more highly associated with preclinical Alzheimer's disease than imaging and cognitive markers of neurodegeneration*. **Alzheimer's & Dementia: Diagnosis, Assessment & Disease Monitoring**, 4, 18–27.
34. Legant, W. R.; Shao, L.; Grimm, J. B.; Brown, T. A.; Milkie, D. E.; **Avants, B. B.**; Lavis, L. D.; Betzig, E. (2016). *High-density three-dimensional localization microscopy across large volumes*. **Nature Methods**, 13(4), 359–365.
35. Pustina, D.; Coslett, H. B.; Turkeltaub, P. E.; Tustison, N. J.; Schwartz, M. F.; **Avants, B. B.** (2016). *Automated segmentation of chronic stroke lesions using LINDA: Lesion identification with neighborhood data analysis*. **Human Brain Mapping**, 37(4), 1405–1421.
36. Betancourt, L. M.; **Avants, B. B.**; Farah, M. J.; Brodsky, N. L.; Wu, J.; Ashtari, M.; Hurt, H. (2016). *Effect of socioeconomic status (SES) disparity on neural development in female African-American infants at age 1 month*. **Developmental Science**, 19(6), 947–956.
37. **Avants, B. B.**; Hackman, D. A.; Betancourt, L. M.; Lawson, G. M.; Hurt, H.; Farah, M. J. (2016). *Relation of Childhood Home Environment to Cortical Thickness in Late Adolescence: Specificity of Experience and Timing*. **PLOS ONE**, 10(10), e0138217.
38. **Avants, B. B.**; Johnson, H. J.; Tustison, N. J. (2015). *Neuroinformatics and the Insight ToolKit*. **Frontiers in Neuroinformatics**, 9, 5.
39. Pustina, D.; **Avants, B. B.**; Sperling, M.; Gorniak, R.; He, X.; Doucet, G.; Barnett, P.; Mintzer, S.; Sharan, A.; Tracy, J. (2015). *Predicting the laterality of temporal lobe epilepsy from PET, MRI, and DTI: A multimodal study*. **NeuroImage: Clinical**, 9, 20–31.
40. **Avants, B. B.**; Duda, J. T.; Kilroy, E.; Krasileva, K.; Jann, K.; Kandel, B. T.; Tustison, N. J.; Yan, L.; Jog, M.; Smith, R.; Wang, Y.; Dapretto, M.; Wang, D. J. J. (2015). *The pediatric template of brain perfusion*. **Scientific Data**, 2, 150003.
41. Wu, J.; Awate, S. P.; Licht, D. J.; Clouchoux, C.; du Plessis, A. J.; **Avants, B. B.**; Vossough, A.; Gee, J. C.; Limperopoulos, C. (2015). *Assessment of MRI-Based Automated Fetal Cerebral Cortical Folding Measures in Prediction of Gestational Age in the Third Trimester*. **AJNR Am J Neuroradiol**, 36(7), 1369–1374.
42. Isgum, I.; Benders, M. J.; **Avants, B.**; Cardoso, M. J.; Counsell, S. J., et al. (2015). *Evaluation of automatic neonatal brain segmentation algorithms: The NeoBrainS12 challenge*. **Medical Image Analysis**, 20(1), 135–151.
43. Kandel, B. M.; Wang, D. J. J.; Detre, J. A.; Gee, J. C.; **Avants, B. B.** (2015). *Decomposing cerebral blood flow MRI into functional and structural components: a non-local approach based on prediction*. **NeuroImage**, 105, 156–170.
44. Kandel, B. M.; **Avants, B. B.**; Gee, J. C.; Arnold, S. E.; Wolk, D. A. (2015). *Neuropsychological Testing Predicts Cerebrospinal Fluid Amyloid-β in Mild Cognitive Impairment*. **Journal of Alzheimer's Disease**, 46(4), 901–912.
45. Kandel, B. M.; Wang, D. J. J.; Gee, J. C.; **Avants, B. B.** (2015). *Eigenanatomy: sparse dimensionality reduction for multi-modal medical image analysis*. **Methods**, 73, 43–53.
46. Adler, D. H.; Pluta, J.; Kadivar, S.; Craige, C.; Gee, J. C.; **Avants, B. B.**; Yushkevich, P. A. (2014). *Histology-derived volumetric annotation of the human hippocampal subfields in postmortem MRI*. **NeuroImage**, 84, 505–523.
47. **Avants, B. B.**; Tustison, N. J.; Stauffer, M.; Song, G.; Wu, B.; Gee, J. C. (2014). *The Insight ToolKit image registration framework*. **Frontiers in Neuroinformatics**, 8, 44.
48. **Avants, B. B.**; Libon, D. J.; Rascovsky, K.; Boller, A.; McMillan, C. T.; Massimo, L.; Coslett, H. B.; Chatterjee, A.; Gross, R. G.; Grossman, M. (2014). *Sparse canonical correlation analysis relates network-level atrophy to multivariate cognitive measures in a neurodegenerative population*. **NeuroImage**, 84, 698–711.
49. Cook, P. A.; McMillan, C. T.; **Avants, B. B.**; Peelle, J. E.; Gee, J. C.; Grossman, M. (2014). *Relating brain anatomy and cognitive ability using a multivariate multimodal framework*. **NeuroImage**, 99, 477–486.
50. Dhillon, P. S.; Wolk, D. A.; Das, S. R.; Ungar, L. H.; Gee, J. C.; **Avants, B. B.** (2014). *Subject-specific functional parcellation via Prior Based Eigenanatomy*. **NeuroImage**, 99, 14–27.
51. McMillan, C. T.; **Avants, B. B.**; Cook, P.; Ungar, L.; Trojanowski, J. Q.; Grossman, M. (2014). *The power of neuroimaging biomarkers for screening frontotemporal dementia*. **Human Brain Mapping**, 35(9), 4827–4840.
52. McMillan, C. T.; Toledo, J. B.; **Avants, B. B.**; Cook, P. A.; Wood, E. M.; Suh, E.; Irwin, D. J.; Powers, J.; Olm, C.; Elman, L.; McCluskey, L.; Schellenberg, G. D.; Lee, V. M.-Y.; Trojanowski, J. Q.; Van Deerlin, V. M.; Grossman, M. (2014). *Genetic and neuroanatomic associations in sporadic frontotemporal lobar degeneration*. **Neurobiology of Aging**, 35(6), 1473–1482.
53. Tustison, N. J.; Cook, P. A.; Klein, A.; Song, G.; Das, S. R.; Duda, J. T.; Kandel, B. M.; van Strien, N.; Stone, J. R.; Gee, J. C.; **Avants, B. B.** (2014). *Large-scale evaluation of ANTs and FreeSurfer cortical thickness measurements*. **NeuroImage**, 99, 166–179.
54. Hopkins, W. D.; **Avants, B. B.** (2013). *Regional and Hemispheric Variation in Cortical Thickness in Chimpanzees (Pan troglodytes)*. **Journal of Neuroscience**, 33(12), 5241–5248.
55. Kim, J.; **Avants, B.**; Whyte, J.; Gee, J. C. (2013). *Methodological considerations in longitudinal morphometry of traumatic brain injury*. **Frontiers in Human Neuroscience**, 7, 52.
56. Lawson, G. M.; Duda, J. T.; **Avants, B. B.**; Wu, J.; Farah, M. J. (2013). *Associations between children's socioeconomic status and prefrontal cortical thickness*. **Developmental Science**, 16(5), 641–652.
57. McMillan, C. T.; **Avants, B.**; Irwin, D. J.; Toledo, J. B.; Wolk, D. A.; Van Deerlin, V. M.; Shaw, L. M.; Trojanowski, J. Q.; Grossman, M. (2013). *Can MRI screen for CSF biomarkers in neurodegenerative disease?*. **Neurology**, 80(2), 132–138.
58. McMillan, C. T.; Irwin, D. J.; **Avants, B. B.**; Powers, J.; Cook, P. A.; Toledo, J. B.; McCarty Wood, E.; Van Deerlin, V. M.; Lee, V. M.-Y.; Trojanowski, J. Q.; Grossman, M. (2013). *White matter imaging helps dissociate tau from TDP-43 in frontotemporal lobar degeneration*. **J. Neurol. Neurosurg. Psychiatry**, 84(9), 949–955.
59. Tustison, N. J.; **Avants, B. B.** (2013). *Explicit B-spline regularization in diffeomorphic image registration*. **Frontiers in Neuroinformatics**, 7, 39.
60. Tustison, N. J.; Johnson, H. J.; Rohlfing, T.; Klein, A.; Ghosh, S. S.; Ibanez, L.; **Avants, B. B.** (2013). *Instrumentation bias in the use and evaluation of scientific software: recommendations for reproducible practices in the computational sciences*. **Frontiers in Neuroscience**, 7, 162.
61. Weber, M. J.; Detre, J. A.; Thompson-Schill, S. L.; **Avants, B. B.** (2013). *Reproducibility of functional network metrics and network structure: a comparison of task-related BOLD, resting ASL with BOLD contrast, and resting cerebral blood flow*. **Cogn. Affect. Behav. Neurosci.**, 13(3), 627–640.
62. Badea, A.; Gewalt, S.; **Avants, B. B.**; Cook, J. J.; Johnson, G. A. (2012). *Quantitative mouse brain phenotyping based on single and multispectral MR protocols*. **NeuroImage**, 63(3), 1633–1645.

63. Cook, P. A.; **Avants, B. B.**; McMillan, C. T.; Powers, J.; Gee, J. C.; Grossman, M. (2012). *Multimodal neuroimaging reveals gray and white matter associations with verbal fluency in frontotemporal degeneration*. **Dement. Geriatr. Cogn. Disord.**, 33(1), 154–155.
64. Das, S. R.; **Avants, B. B.**; Pluta, J.; Wang, H.; Suh, J. W.; Weiner, M. W.; Mueller, S. G.; Yushkevich, P. A. (2012). *Measuring longitudinal change in the hippocampal formation from in vivo high-resolution T2-weighted MRI*. **NeuroImage**, 60(2), 1266–1279.
65. Datta, R.; Lee, J.; Duda, J.; **Avants, B. B.**; Vite, C. H.; Tseng, B.; Gee, J. C.; Aguirre, G. D.; Aguirre, G. K. (2012). *A Digital Atlas of the Dog Brain*. **PLOS ONE**, 7(12), e52140.
66. Ghosh, S. S.; Klein, A.; **Avants, B.**; Millman, K. J. (2012). *Learning from open source software projects to improve scientific review*. **Frontiers in Computational Neuroscience**, 6, 18.
67. Gross, R. G.; McMillan, C. T.; Chandrasekaran, K.; Dreyfuss, M.; Ash, S.; **Avants, B.**; Cook, P.; Moore, P.; Libon, D. J.; Siderowf, A.; Grossman, M. (2012). *Sentence processing in Lewy body spectrum disorder: The role of working memory*. **Brain and Cognition**, 78(2), 85–93.
68. Hanson, J. L.; Chung, M. K.; **Avants, B. B.**; Rudolph, K. D.; Shirtcliff, E. A.; Gee, J. C.; Davidson, R. J.; Pollak, S. D. (2012). *Structural Variations in Prefrontal Cortex Mediate the Relationship between Early Childhood Stress and Spatial Working Memory*. **Journal of Neuroscience**, 32(23), 7917–7925.
69. Hanson, J. L.; Suh, J. W.; Nacewicz, B. M.; Sutterer, M. J.; Cayo, A. A.; Stodola, D. E.; Burghy, C. A.; Wang, H.; **Avants, B. B.**; Yushkevich, P. A.; Essex, M. J.; Pollak, S. D.; Davidson, R. J. (2012). *Robust Automated Amygdala Segmentation via Multi-Atlas Diffeomorphic Registration*. **Frontiers in Neuroscience**, 6, 166.
70. Hurst, D. R.; Schoenemann, P. T.; Loyet, M. M.; **Avants, B. B.**; Gee, J. C. (2012). *How well does endocranial morphology predict behavioral differences in primates?*. **Am. J. Phys. Anthropol.**, 147(54), 171.
71. Jain, V.; Duda, J.; **Avants, B.**; Giannetta, M.; Xie, S. X.; Roberts, T.; Detre, J. A.; Hurt, H.; Wehrli, F. W.; Wang, D. J. J. (2012). *Longitudinal Reproducibility and Accuracy of Pseudo-Continuous Arterial Spin-labeled Perfusion MR Imaging in Typically Developing Children*. **Radiology**, 263(2), 527–536.
72. Libon, D. J.; McMillan, C.; **Avants, B.**; Boller, A.; Morgan, B.; Burkholder, L.; Chandrasekaran, K.; Elman, L.; McCluskey, L.; Grossman, M. (2012). *Deficits in Concept Formation in Amyotrophic Lateral Sclerosis*. **Neuropsychology**, 26(4), 422–429.
73. Loyet, M. M.; Schoenemann, P. T.; **Avants, B. B.**; Gee, J. C. (2012). *Associations between localized variation in brain anatomy and social behavior in healthy human subjects*. **Am. J. Phys. Anthropol.**, 147(54), 196.
74. Rohlfing, T.; **Avants, B.** (2012). *"Nonparametric Local Smoothing" is not image registration*. **BMC Research Notes**, 5, 610.
75. Tustison, N. J.; **Avants, B. B.**; Cook, P. A.; Kim, J.; Whyte, J.; Gee, J. C.; Stone, J. R. (2012). *Logical circularity in voxel-based analysis: Normalization strategy may induce statistical bias*. **Human Brain Mapping**, 35(3), 745–759.
76. Ashtari, M.; **Avants, B.**; Cyckowski, L.; Cervellione, K. L.; Roofeh, D.; Cook, P.; Gee, J.; Sevy, S.; Kumra, S. (2011). *Medial temporal structures and memory functions in adolescents with heavy cannabis use*. **J. Psychiatr. Res.**, 45(8), 1055–1066.
77. **Avants, B. B.**; Tustison, N. J.; Song, G.; Cook, P. A.; Klein, A.; Gee, J. C. (2011). *A reproducible evaluation of ANTs similarity metric performance in brain image registration*. **NeuroImage**, 54(3), 2033–2044.
78. **Avants, B. B.**; Tustison, N. J.; Wu, J.; Cook, P. A.; Gee, J. C. (2011). *An Open Source Multivariate Framework for n-Tissue Segmentation with Evaluation on Public Data*. **Neuroinformatics**, 9(4), 381–400.
79. Boller, A.; Libon, D.; Rascovsky, K.; Gross, R. G.; Dreyfuss, M.; **Avants, B.**; Massimo, L.; Moore, P.; Kitain, J.; Coslett, H.; Chatterjee, A.; Grossman, M. (2011). *Philadelphia Brief Assessment of Cognition (PBAC): A Validated Screening Measure for Dementia*. **Neurology**, 76(9, 4), A511.
80. Das, S. R.; Mechanic-Hamilton, D.; Pluta, J.; Korczykowski, M.; Detre, J. A.; Yushkevich, P. A. (2011). *Heterogeneity of functional activation during memory encoding across hippocampal subfields in temporal lobe epilepsy*. **NeuroImage**, 58(4), 1121–1130.
81. Morgan, B.; Gross, R. G.; Clark, R.; Dreyfuss, M.; Boller, A.; Camp, E.; Liang, T.-W.; **Avants, B.**; McMillan, C. T.; Grossman, M. (2011). *Some is not enough: Quantifier comprehension in corticobasal syndrome and behavioral variant frontotemporal dementia*. **Neuropsychologia**, 49(13), 3532–3541.
82. Murphy, K.; van Ginneken, B.; Reinhardt, J. M., ..., Tustison, N.; **Avants, B.**; Gee, J. C., ..., Pluim, J. P. W. (2011). *Evaluation of Registration Methods on Thoracic CT: The EMPIRE10 Challenge*. **IEEE Transactions on Medical Imaging**, 30(11), 1901–1920.
83. Schoenemann, P. T.; Holloway, R.; Monge, J.; **Avants, B.**; Gee, J. (2011). *Differences in endocranial shape between Homo and Pongids assessed through non-rigid deformation analysis of high-resolution CT images*. **Am. J. Phys. Anthropol.**, 144(52), 265–266.
84. Tustison, N.; **Avants, B.**; Cook, P.; Kim, J.; Whyte, J.; Gee, J.; Ahlers, S.; Stone, J. (2011). *Multivariate analysis of diffusion tensor imaging and cortical thickness maps in a traumatic brain injury (TBI) cohort using Advanced Normalization Tools (ANTs)*. **J. Neurotrauma**, 28(6), A111.
85. Tustison, N. J.; **Avants, B. B.**; Flors, L.; Altes, T. A.; de Lange, E. E.; Mugler III, J. P.; Gee, J. C. (2011). *Ventilation-Based Segmentation of the Lungs Using Hyperpolarized He-3 MRI*. **J. Magn. Reson. Imaging**, 34(4), 831–841.
86. Tustison, N. J.; **Avants, B. B.**; Siqueira, M.; Gee, J. C. (2011). *Topological Well-Composedness and Glamorous Glue: A Digital Gluing Algorithm for Topologically Constrained Front Propagation*. **IEEE Trans Image Process**, 20(6), 1756–1761.
87. Wang, H.; Das, S. R.; Suh, J. W.; Altinay, M.; Pluta, J.; Craige, C.; **Avants, B.**; Yushkevich, P. A.; ADNI (2011). *A learning-based wrapper method to correct systematic errors in automatic image segmentation: Consistently improved performance in hippocampus, cortex and brain segmentation*. **NeuroImage**, 55(3), 968–985.
88. Ash, S.; McMillan, C.; Gunawardena, D.; **Avants, B.**; Morgan, B.; Khan, A.; Moore, P.; Gee, J.; Grossman, M. (2010). *Speech errors in progressive non-fluent aphasia*. **Brain and Language**, 113(1), 13–20.
89. **Avants, B. B.**; Cook, P. A.; Ungar, L.; Gee, J. C.; Grossman, M. (2010). *Dementia induces correlated reductions in white matter integrity and cortical thickness: A multivariate neuroimaging study with sparse canonical correlation analysis*. **NeuroImage**, 50(3), 1004–1016.
90. **Avants, B. B.**; Yushkevich, P.; Pluta, J.; Minkoff, D.; Korczykowski, M.; Detre, J.; Gee, J. C. (2010). *The optimal template effect in hippocampus studies of diseased populations*. **NeuroImage**, 49(3), 2457–2466.
91. Farag, C.; Troiani, V.; Bonner, M.; Powers, C.; **Avants, B.**; Gee, J.; Grossman, M. (2010). *Hierarchical Organization of Scripts: Converging Evidence from fMRI and Frontotemporal Degeneration*. **Cerebral Cortex**, 20(10), 2453–2463.
92. Grossman, M.; Eslinger, P. J.; Troiani, V.; Anderson, C.; **Avants, B.**; Gee, J. C.; McMillan, C.; Massimo, L.; Khan, A.; Antani, S. (2010). *The role of ventral medial prefrontal cortex in social decisions: Converging evidence from fMRI and frontotemporal lobar degeneration*. **Neuropsychologia**, 48(12), 3505–3512.
93. Gunawardena, D.; Ash, S.; McMillan, C.; **Avants, B.**; Gee, J.; Grossman, M. (2010). *Why are patients with progressive nonfluent aphasia nonfluent?*.

Neurology, 75(7), 588–594.

94. Hanson, J. L.; Chung, M. K.; **Avants, B. B.**; Shirtcliff, E. A.; Gee, J. C.; Davidson, R. J.; Pollak, S. D. (2010). *Early Stress Is Associated with Alterations in the Orbitofrontal Cortex: A Tensor-Based Morphometry Investigation of Brain Structure and Behavioral Risk*. **Journal of Neuroscience**, 30(22), 7466–7472.
95. Kim, J.; Whyte, J.; Patel, S.; **Avants, B.**; Europa, E.; Wang, J.; Slattery, J.; Gee, J. C.; Coslett, H. B.; Detre, J. A. (2010). *Resting Cerebral Blood Flow Alterations in Chronic Traumatic Brain Injury: An Arterial Spin Labeling Perfusion fMRI Study*. **J. Neurotrauma**, 27(8), 1399–1411.
96. Klein, A.; Ghosh, S. S.; **Avants, B.**; Yeo, B. T. T.; Fischl, B.; Ardekani, B.; Gee, J. C.; Mann, J. J.; Parsey, R. V. (2010). *Evaluation of volume-based and surface-based brain image registration methods*. **NeuroImage**, 51(1), 214–220.
97. Rao, H.; Betancourt, L.; Giannetta, J. M.; Brodsky, N. L.; Korczykowski, M.; **Avants, B. B.**; Gee, J. C.; Wang, J.; Hurt, H.; Detre, J. A.; Farah, M. J. (2010). *Early parental care is important for hippocampal maturation: Evidence from brain morphology in humans*. **NeuroImage**, 49(1), 1144–1150.
98. Schoenemann, P. T.; Monge, J.; Holloway, R. L.; **Avants, B. B.**; Gee, J. C. (2010). *Creating statistical atlases of modern primate endocranial morphology using non-rigid deformation analysis of high-resolution CT images*. **Am. J. Phys. Anthropol.**, (50), 208–209.
99. Tustison, N. J.; **Avants, B. B.**; Cook, P. A.; Zheng, Y.; Egan, A.; Yushkevich, P. A.; Gee, J. C. (2010). *N4ITK: Improved N3 bias correction*. **IEEE Transactions on Medical Imaging**, 29(6), 1310–1320.
100. Wang, D. J. J.; Bi, X.; **Avants, B. B.**; Meng, T.; Zuehlsdorff, S.; Detre, J. A. (2010). *Estimation of Perfusion and Arterial Transit Time in Myocardium Using Free-Breathing Myocardial Arterial Spin Labeling With Navigator-Echo*. **Magn. Reson. Med.**, 64(5), 1289–1295.
101. Yushkevich, P. A.; **Avants, B. B.**; Das, S. R.; Pluta, J.; Altinay, M.; Craige, C.; ADNI (2010). *Bias in estimation of hippocampal atrophy using deformation-based morphometry arises from asymmetric global normalization: An illustration in ADNI 3T MRI data*. **NeuroImage**, 50(2), 434–445.
102. Yushkevich, P. A.; Wang, H.; Pluta, J.; Das, S. R.; Craige, C.; **Avants, B. B.**; Weiner, M. W.; Mueller, S. (2010). *Nearly automatic segmentation of hippocampal subfields in in vivo focal T2-weighted MRI*. **NeuroImage**, 53(4), 1208–1224.
103. Ash, S.; Moore, P.; Vesely, L.; Gunawardena, D.; McMillan, C.; Anderson, C.; **Avants, B.**; Grossman, M. (2009). *Non-fluent speech in frontotemporal lobar degeneration*. **Journal of Neurolinguistics**, 22(4), 370–383.
104. **Avants, B.**; Khan, A.; McCluskey, L.; Elman, L.; Grossman, M. (2009). *Longitudinal Cortical Atrophy in Amyotrophic Lateral Sclerosis With Frontotemporal Dementia*. **Archives of Neurology**, 66(1), 138–139.
105. Bonner, M. F.; Vesely, L.; Price, C.; Anderson, C.; Richmond, L.; Farag, C.; **Avants, B.**; Grossman, M. (2009). *Reversal of the concreteness effect in semantic dementia*. **Cognitive Neuropsychology**, 26(6), 568–579.
106. Das, S. R.; **Avants, B. B.**; Grossman, M.; Gee, J. C. (2009). *Registration based cortical thickness measurement*. **NeuroImage**, 45(3), 867–879.
107. Das, S. R.; Mechanic-Hamilton, D.; Korczykowski, M.; Pluta, J.; Glynn, S.; **Avants, B. B.**; Detre, J. A.; Yushkevich, P. A. (2009). *Structure Specific Analysis of the Hippocampus in Temporal Lobe Epilepsy*. **Hippocampus**, 19(6), 517–525.
108. Klein, A.; Andersson, J.; Ardekani, B. A.; Ashburner, J.; **Avants, B.**, et al. (2009). *Evaluation of 14 nonlinear deformation algorithms applied to human brain MRI registration*. **NeuroImage**, 46(3), 786–802.
109. Massimo, L.; Powers, C.; Moore, P.; Vesely, L.; **Avants, B.**; Gee, J.; Libon, D. J.; Grossman, M. (2009). *Neuroanatomy of Apathy and Disinhibition in Frontotemporal Lobar Degeneration*. **Dement. Geriatr. Cogn. Disord.**, 27(1), 96–104.
110. Pluta, J.; **Avants, B. B.**; Glynn, S.; Awate, S.; Gee, J. C.; Detre, J. A. (2009). *Appearance and Incomplete Label Matching for Diffeomorphic Template Based Hippocampus Segmentation*. **Hippocampus**, 19(6), 565–571.
111. Schoenemann, P. T.; Monge, J.; **Avants, B. B.**; Gee, J. C. (2009). *An atlas of modern human cranial morphology constructed via non-rigid deformation analysis of high-resolution CT images*. **Am. J. Phys. Anthropol.**, 231.
112. Tustison, N. J.; **Avants, B. B.**; Gee, J. C. (2009). *Directly Manipulated Free-Form Deformation Image Registration*. **IEEE Trans Image Process**, 18(3), 624–635.
113. Yushkevich, P. A.; **Avants, B. B.**; Pluta, J.; Das, S.; Minkoff, D.; Mechanic-Hamilton, D.; Glynn, S.; Pickup, S.; Liu, W.; Gee, J. C.; Grossman, M.; Detre, J. A. (2009). *A high-resolution computational atlas of the human hippocampus from postmortem magnetic resonance imaging at 9.4 T*. **NeuroImage**, 44(2), 385–398.
114. **Avants, B.**; Duda, J. T.; Kim, J.; Zhang, H.; Pluta, J.; Gee, J. C.; Whyte, J. (2008). *Multivariate Analysis of Structural and Diffusion Imaging in Traumatic Brain Injury*. **Academic Radiology**, 15(11), 1360–1375.
115. **Avants, B. B.**; Epstein, C. L.; Grossman, M.; Gee, J. C. (2008). *Symmetric diffeomorphic image registration with cross-correlation: Evaluating automated labeling of elderly and neurodegenerative brain*. **Medical Image Analysis**, 12(1), 26–41.
116. Grossman, M.; Anderson, C.; Khan, A.; **Avants, B.**; Elman, L.; McCluskey, L. (2008). *Impaired action knowledge in amyotrophic lateral sclerosis*. **Neurology**, 71(18), 1396–1401.
117. Grossman, M.; Anderson, C.; Khan, A.; **Avants, B.**; Elman, L.; McCluskey, L. (2008). *Neural basis for impaired action knowledge in amyotrophic lateral sclerosis*. **Neurology**, 70(11, 1), A248.
118. Kim, J.; **Avants, B.**; Patel, S.; Whyte, J.; Coslett, B. H.; Pluta, J.; Detre, J. A.; Gee, J. C. (2008). *Structural consequences of diffuse traumatic brain injury: A large deformation tensor-based morphometry study*. **NeuroImage**, 39(3), 1014–1026.
119. Massimo, L. M.; Anderson, C.; Moore, P.; **Avants, B.**; Libon, D.; Cynwyd, B.; Grossman, M. (2008). *Neuroanatomical correlates of apathy and disinhibition in frontotemporal dementia*. **Neurology**, 70(11, 1), A443.
120. Schoenemann, P. T.; Holloway, R. L.; **Avants, B. B.**; Gee, J. C. (2008). *Endocast asymmetry in pongids assessed via non-rigid deformation analysis of high-resolution CT images*. **Am. J. Phys. Anthropol.**, (46), 188.
121. Schoenemann, P. T.; Holloway, R. L.; **Avants, B. B.**; Gee, J. C. (2008). *The role of micro-morphological stress markers in the differential diagnosis of infectious bone diseases*. **Am. J. Phys. Anthropol.**, (46), 188–189.
122. Simon, T. J.; Wu, Z.; **Avants, B.**; Zhang, H.; Gee, J. C.; Stebbins, G. T. (2008). *Atypical cortical connectivity and visuospatial cognitive impairments are related in children with chromosome 22q11.2 deletion syndrome*. **Behav Brain Funct**, 4, 25.
123. Aguirre, G. K.; Komaromy, A. M.; Cideciyan, A. V.; Brainard, D. H.; Aleman, T. S.; Roman, A. J.; **Avants, B. B.**; Gee, J. C.; Korczykowski, M.; Hauswirth, W. W.; Acland, G. M.; Aguirre, G. D.; Jacobson, S. G. (2007). *Canine and human visual cortex intact and responsive despite early retinal blindness from RPE65 mutation*. **PLoS Medicine**, 4(6), 1117–1128.
124. **Avants, B. B.**; Hurt, H.; Giannetta, J. M.; Epstein, C. L.; Shera, D. M.; Rao, H.; Wang, J.; Gee, J. C. (2007). *Effects of heavy in utero cocaine exposure on adolescent caudate morphology*. **Pediatric Neurology**, 37(4), 275–279.
125. Fan, Y.; Rao, H.; Hurt, H.; Giannetta, J.; Korczykowski, M.; Shera, D.; **Avants, B. B.**; Gee, J. C.; Wang, J.; Shen, D. (2007). *Multivariate examination*

of brain abnormality using both structural and functional MRI. **NeuroImage**, 36(4), 1189–1199.

126. Ng, L.; Pathak, S. D.; Kuan, C.; Lau, C.; Dong, H.; Sodt, A.; Dang, C.; **Avants, B.**; Yushkevich, P.; Gee, J. C.; Haynor, D.; Lein, E.; Jones, A.; Hawrylycz, M. (2007). *Neuroinformatics for genome-wide 3D gene expression mapping in the mouse brain*. **IEEE/ACM Trans. Comput. Biol. Bioinform.**, 4(3), 382–393.
127. Rao, H.; Wang, J.; Giannetta, J.; Korczykowski, M.; Shera, D.; **Avants, B. B.**; Gee, J.; Detre, J. A.; Hurt, H. (2007). *Altered resting cerebral blood flow in adolescents with in utero cocaine exposure revealed by perfusion functional MRI*. **Pediatrics**, 120(5), E1245–E1254.
128. Schoenemann, P. T.; Gee, J.; **Avants, B.**; Holloway, R. L.; Monge, J.; Lewis, J. (2007). *Validation of plaster endocast morphology through 3D CT image analysis*. **Am. J. Phys. Anthropol.**, 132(2), 183–192.
129. Schoenemann, P. T.; Monge, J.; **Avants, B. B.**; Glotzer, D.; Gee, J. C. (2007). *Sex differences in cranial form assessed via non-rigid deformation analysis of high-resolution CT images*. **Am. J. Phys. Anthropol.**, (44), 209.
130. Zhang, H.; **Avants, B. B.**; Yushkevich, P. A.; Woo, J. H.; Wang, S.; McCluskey, L. F.; Elman, L. B.; Melhem, E. R.; Gee, J. C. (2007). *High-dimensional spatial normalization of diffusion tensor images improves the detection of white matter differences: an example study using amyotrophic lateral sclerosis*. **IEEE Transactions on Medical Imaging**, 26(11), 1585–1597.
131. **Avants, B. B.**; Schoenemann, P. T.; Gee, J. C. (2006). *Lagrangian frame diffeomorphic image registration: Morphometric comparison of human and chimpanzee cortex*. **Medical Image Analysis**, 10(3), 397–412.
132. **Avants, B.**; Grossman, M.; Gee, J. C. (2005). *The correlation of cognitive decline with frontotemporal dementia induced annualized gray matter loss using diffeomorphic morphometry*. **Alzheimer Dis. Assoc. Disord.**, 19 Suppl 1, S25–S28.
133. **Avants, B.**; Gee, J.; Schoenemann, P.; Monge, J.; Lewis, J.; Holloway, R. (2005). *A new method for assessing endocast morphology: calculating local curvature from 3D CT images*. **Am. J. Phys. Anthropol.**, (40), 67.
134. Sundaram, T. A.; **Avants, B. B.**; Gee, J. C. (2005). *Towards a dynamic model of pulmonary parenchymal deformation: evaluation of methods for temporal reparameterization of lung data*. **MICCAI**, 8(Pt 2), 328–335.
135. **Avants, B.**; Gee, J. (2004). *Geodesic estimation for large deformation anatomical shape averaging and interpolation*. **NeuroImage**, 23(1), S139–S150.
136. **Avants, B.**; Gee, J.; Schoenemann, P. T.; Monge, J.; Lewis, J. E.; Holloway, R. L. (2004). *Validation of plaster endocast morphology through 3D CT image analysis*. **Am. J. Phys. Anthropol.**, (38), 56.
137. Schoenemann, P. T.; **Avants, B. B.**; Gee, J. C.; Glotzer, L. D.; Sheehan, M. J. (2004). *Analysis of chimp-human brain differences via non-rigid deformation of 3D MR images*. **Am. J. Phys. Anthropol.**, (38), 174–175.
138. **Avants, B.**; Gee, J. (2003). *The shape operator for differential analysis of images*. **Information Processing in Medical Imaging (IPMI)**, 18, 101–113.
139. Dubb, A.; Gur, R.; **Avants, B.**; Gee, J. (2003). *Characterization of sexual dimorphism in the human corpus callosum*. **NeuroImage**, 20(1), 512–519.
140. **Avants, B.**; Soodak, D.; Ruppeiner, G. (1999). *Measuring the electrical conductivity of the earth*. **American Journal of Physics**, 67(7), 593–598.

SELECTED PEER-REVIEWED CONFERENCE PUBLICATIONS

1. Wu, Z.; Dadu, A.; Tustison, N.; **Avants, B.**; Nalls, M.; Sun, J.; Faghri, F. (2023). *Multimodal Patient Representation Learning with Missing Modalities and Labels*. **ICLR 2023**.
2. Hutchison, R. M.; Chiao, P.; **Avants, B. B.** (2017). *Decomposing the interdependence of brain anatomy, connectivity, and perfusion*. **OHBM Annual Meeting**, Vancouver, Canada.
3. Pustina, D.; Coslett, B.; Schwartz, M. F.; **Avants, B. B.** (2015). *A supervised framework for lesion segmentation and automated VLSM analyses in left hemispheric stroke*. **Academy of Aphasia 53rd Annual Meeting**, Tucson, AZ.
4. Kandel, B. M.; Wang, D. J. J.; Gee, J. C.; **Avants, B. B.** (2014). *Single-subject structural networks with closed-form rotation invariant matching improve power in developmental studies of the cortex*. **MICCAI**, 17(Pt 3), 137–144.
5. Xie, L.; Pluta, J.; Wang, H.; Das, S. R.; Mancuso, L.; Kliot, D.; **Avants, B. B.**; Ding, S.-L.; Wolk, D. A.; Yushkevich, P. A. (2014). *Automatic clustering and thickness measurement of anatomical variants of the human perirhinal cortex*. **MICCAI**, 17(Pt 3), 81–88.
6. Duda, J. T.; Detre, J. A.; Kim, J.; Gee, J. C.; **Avants, B. B.** (2013). *Fusing functional signals by sparse canonical correlation analysis improves network reproducibility*. **MICCAI**, 16(Pt 3), 635–642.
7. Dhillon, P.; Kandel, B. M.; Wolk, D.; Gee, J. C.; **Avants, B.** (2013). *Prior-based Eigenanatomy for Classification of Autism Spectrum Disorder*. **PRNI 2013**.
8. Kandel, B. M.; Wolk, D.; Gee, J. C.; **Avants, B.** (2013). *Predicting Cognitive Data From Medical Images Using Sparse Linear Regression*. **Information Processing in Medical Imaging (IPMI)**.
9. **Avants, B.**; Dhillon, P.; Kandel, B. M.; Cook, P. A.; McMillan, C. T.; Grossman, M.; Gee, J. C. (2012). *Eigenanatomy improves detection power for longitudinal cortical change*. **MICCAI**, 15, 206–213.
10. **Avants, B.**; Cook, P. A.; McMillan, C.; Grossman, M.; Tustison, N. J.; Zheng, Y.; Gee, J. C. (2010). *Sparse unbiased analysis of anatomical variance in longitudinal imaging*. **MICCAI**, 13, 324–331.
11. Wang, H.; Das, S.; Pluta, J.; Craige, C.; Altinay, M.; **Avants, B.**; Weiner, M.; Mueller, S.; Yushkevich, P. (2010). *Standing on the shoulders of giants: improving medical image segmentation via bias correction*. **MICCAI**, 13, 105–112.
12. Sun, H.; **Avants, B. B.**; Frangi, A. F.; Sukno, F.; Gee, J. C.; Yushkevich, P. A. (2008). *Cardiac medial modeling and time-course heart wall thickness analysis*. **MICCAI**, 11, 766–773.
13. Yushkevich, P. A.; **Avants, B. B.**; Pluta, J.; Minkoff, D.; Detre, J. A.; Grossman, M.; Gee, J. C. (2008). *Shape-based alignment of hippocampal subfields: evaluation in postmortem MRI*. **MICCAI**, 11, 510–517.
14. **Avants, B.**; Anderson, C.; Grossman, M.; Gee, J. C. (2007). *Spatiotemporal normalization for longitudinal analysis of gray matter atrophy in frontotemporal dementia*. **MICCAI**, 10, 303–310.
15. **Avants, B. B.**; Grossman, M.; Gee, J. C. (2006). *Symmetric diffeomorphic image registration: Evaluating automated labeling of elderly and neurodegenerative cortex and frontal lobe*. **WBIR** / **LNCS**, 4057, 50–57.
16. **Avants, B. B.**; Epstein, C. L.; Gee, J. C. (2006). *Geodesic image normalization and temporal parameterization in the space of diffeomorphisms*. **LNCS**, 4091, 9–16.
17. Song, Z.; Tustison, N.; **Avants, B.**; Gee, J. C. (2006). *Integrated graph cuts for brain MRI segmentation*. **MICCAI**, 9, 831–838.

18. Tustison, N. J.; **Avants, B. B.**; Sundaram, T. A.; Duda, J. T.; Gee, J. C. (2006). *A generalization of Free-Form Deformation image registration within the ITK finite element framework*. **WBIR** / **LNCS**, 4057, 238–246.
19. Yushkevich, P. A.; **Avants, B. B.**; Ng, L.; Hawrylycz, M.; Burstein, P. D.; Zhang, H.; Gee, J. C. (2006). *3D mouse brain reconstruction from histology using a coarse-to-fine approach*. **WBIR** / **LNCS**, 4057, 230–237.
20. **Avants, B.**; Epstein, C.; Gee, J. (2005). *Geodesic image interpolation: Parameterizing and interpolating spatiotemporal images*. **LNCS**, 3752, 247–258.
21. Cook, P.; Zhang, H.; **Avants, B.**; Yushkevich, P.; Alexander, D.; Gee, J.; Ciccarelli, O.; Thompson, A. (2005). *An automated approach to connectivity-based partitioning of brain structures*. **MICCAI** / **LNCS**, 3749, 164–171.
22. **Avants, B.**; Gee, J. (2004). *Symmetric geodesic shape averaging and shape interpolation*. **ECCV** / **LNCS**, 3117, 99–110.
23. **Avants, B.**; Gee, J. (2003). *Formulation and evaluation of variational curve matching with prior constraints*. **WBIR** / **LNCS**, 2717, 21–30.
24. **Avants, B.**; Gee, J. (2003). *Continuous curve matching with scale-space curvature and extrema-based scale selection*. **Scale Space** / **LNCS**, 2695, 798–813.
25. **Avants, B.**; Williams, J. (2000). *An adaptive minimal path generation technique for vessel tracking in CTA/CE-MRA volume images*. **MICCAI** / **LNCS**, 1935, 707–716.

BOOK CHAPTERS

1. **Avants, B. B.**; Tustison, N. J. (2022). *Mapping the Spatial Distribution of Lesions in Stroke: Effect of Diffeomorphic Registration Strategy in the ATLAS Dataset*. In **Lesion-to-Symptom Mapping: Principles and Tools** (pp. 79–94). Springer.
2. **Avants, B.** (2018). *Relating high-dimensional structural networks to resting functional connectivity with sparse canonical correlation analysis for neuroimaging*. In **Springer Protocols on Brain Morphometry**.
3. Cullen, N.; **Avants, B.** (2018). *Deep learning for brain segmentation*. In **Springer Protocols on Brain Morphometry**.
4. Gee, J. C.; Zhang, H.; Dubb, A.; **Avants, B.**; Yushkevich, P.; Duda, J. T. (2005). *Anatomy-based visualizations of diffusion tensor images of brain white matter*. In **Visualization and Processing of Tensor Fields**. Springer.
5. **Avants, B.**; Gee, J. C. (2004). *Nonrigid Registration Chapter*. In **Insight Into Images: Principles and Practice for Segmentation, Registration and Image Analysis**. A. K. Peters Ltd., Natick, MA.